Technology for more than forty years, combining his far-ranging insights into the cosmos with a colorful means of expression and an impressive ability to antagonize his colleagues.

Zwicky studied the movement of individual galaxies within a titanic cluster of them, located far beyond the local stars of the Milky Way that trace out the constellation Coma Berenices (the "hair of Berenice," an Egyptian queen in antiquity). The Coma cluster, as we call it, is an isolated and richly populated ensemble of galaxies about 300 million light-years from Earth. Its thousand galaxies orbit the cluster's center, moving in all directions like bees swarming a beehive. Using the motions of a few dozen galaxies as tracers of the gravity field that binds the entire cluster, Zwicky discovered that their average velocity had a shockingly high value. Since larger gravitational forces induce higher velocities in the objects they attract, Zwicky inferred an enormous mass for the Coma cluster. As a reality check on that estimate, you can sum up the masses of each member galaxy that you see. And even though Coma ranks among the largest and most massive clusters in the universe, it does not contain enough visible galaxies to account for the observed speeds Zwicky measured.

How bad is the situation? Have our known laws of gravity failed us? They certainly work within the solar system. Newton showed that you can derive the unique speed that a planet must have to maintain a stable orbit at any distance from the Sun, lest it descend back toward the Sun or ascend to a farther orbit. Turns out, if we could boost Earth's orbital speed to more than the square root of two (1.4142 . . .) times its current value, our planet would achieve "escape velocity," and leave the solar system entirely. We can apply the same reasoning to much larger systems, such as our own Milky Way galaxy, in which stars move in orbits that respond to the gravity from all the other stars; or in clusters of galaxies, where each galaxy likewise feels the gravity from all the other galaxies. In this spirit, amid a page of formulas in his notebook, Einstein wrote a rhyme (more ringingly in German than in this English translation) in honor of Isaac Newton:

> *Look unto the stars to teach us*
> *How the master's thoughts can reach us*
> *Each one follows Newton's math*
> *Silently along its path.*[†]